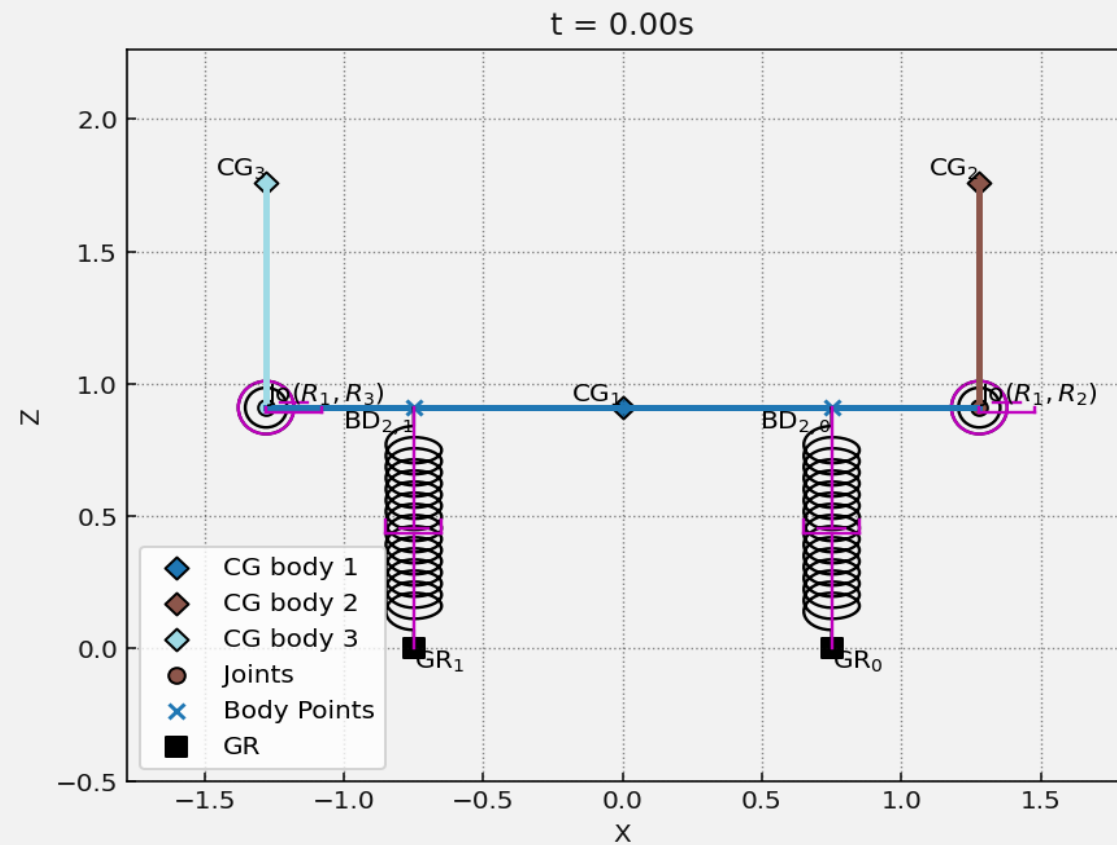
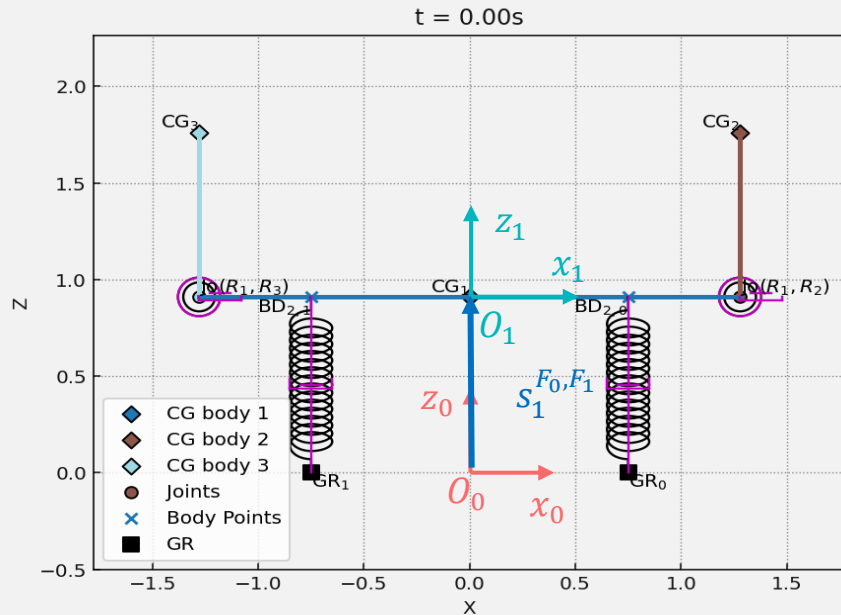


FOSWEC



Bodies definition: body 1



Connecting bodies

Joint type

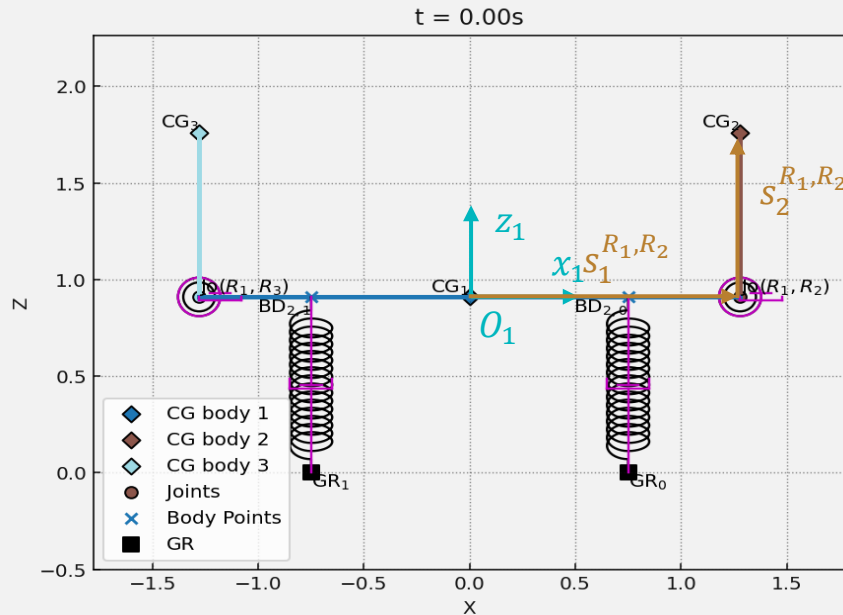
```
joints = [[0, 1], [1, 2], [1, 3]]
types = ['F', 'R', 'R']
parent_cg_to_joint = [[0, 0.91], [2*0.64, 0], [-2*0.64, 0]]
joint_to_child_cg = [[np.nan, np.nan], [0, 5*0.17], [0, 5*0.17]]
prismatic_direction = [[np.nan, np.nan], [np.nan, np.nan], [np.nan, np.nan]]
prismatic_direction = normalize_prismatic_directions(prismatic_direction)
```

$S_0^{F_0, F_1}$
'F' joints only
Parent cg to joint

Direction of
allowed translation

Go to next slide to continue animation

Bodies definition: body 2



Connecting bodies

Joint type

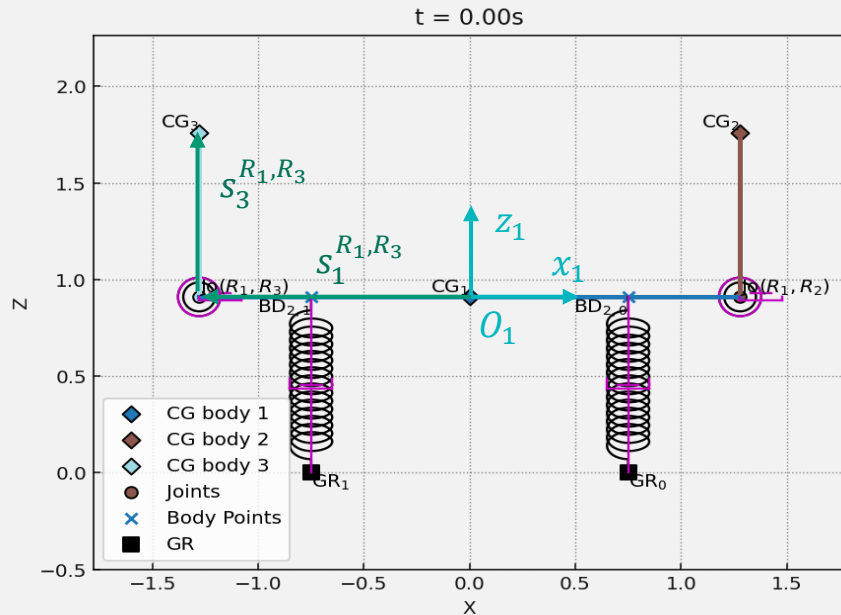
```
joints = [[0, 1], [1, 2], [1, 3]]
types = ['F', 'R', 'R']
parent_cg_to_joint = [[0, 0.91], [2*0.64, 0], [-2*0.64, 0]]
joint_to_child_cg = [[np.nan, np.nan], [0, 5*0.17], [0, 5*0.17]]
prismatic_direction = [[np.nan, np.nan], [np.nan, np.nan], [np.nan, np.nan]]
prismatic_direction = normalize_prismatic_directions(prismatic_direction)
```

$S_1^{R_1, R_2}$

$S_2^{R_1, R_2}$

Go to next slide to continue animation

Bodies definition: body 3



Connecting bodies

Joint type

```

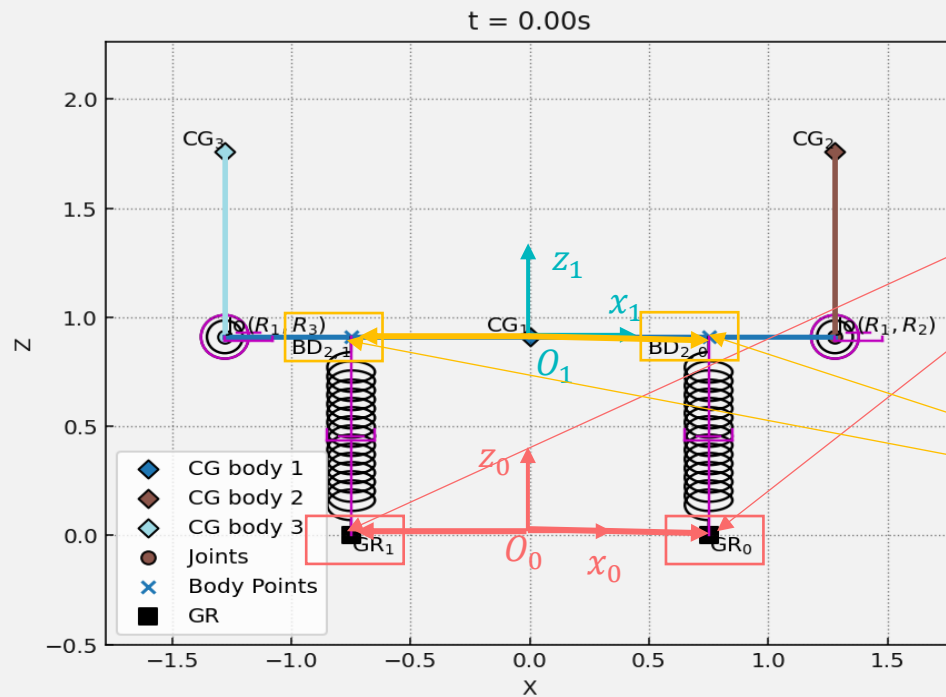
joints = [[0, 1], [1, 2], [1, 3]]
types = ['F', 'R', 'R']
parent_cg_to_joint = [[0, 0.91], [2*0.64, 0], [-2*0.64, 0]]
joint_to_child_cg = [[np.nan, np.nan], [0, 5*0.17], [0, 5*0.17]]
prismatic_direction = [[np.nan, np.nan], [np.nan, np.nan], [np.nan, np.nan]]
prismatic_direction = normalize_prismatic_directions(prismatic_direction)
    
```

$S_1^{R_1, R_3}$

$S_3^{R_1, R_3}$

Go to next slide to continue animation

Points definition

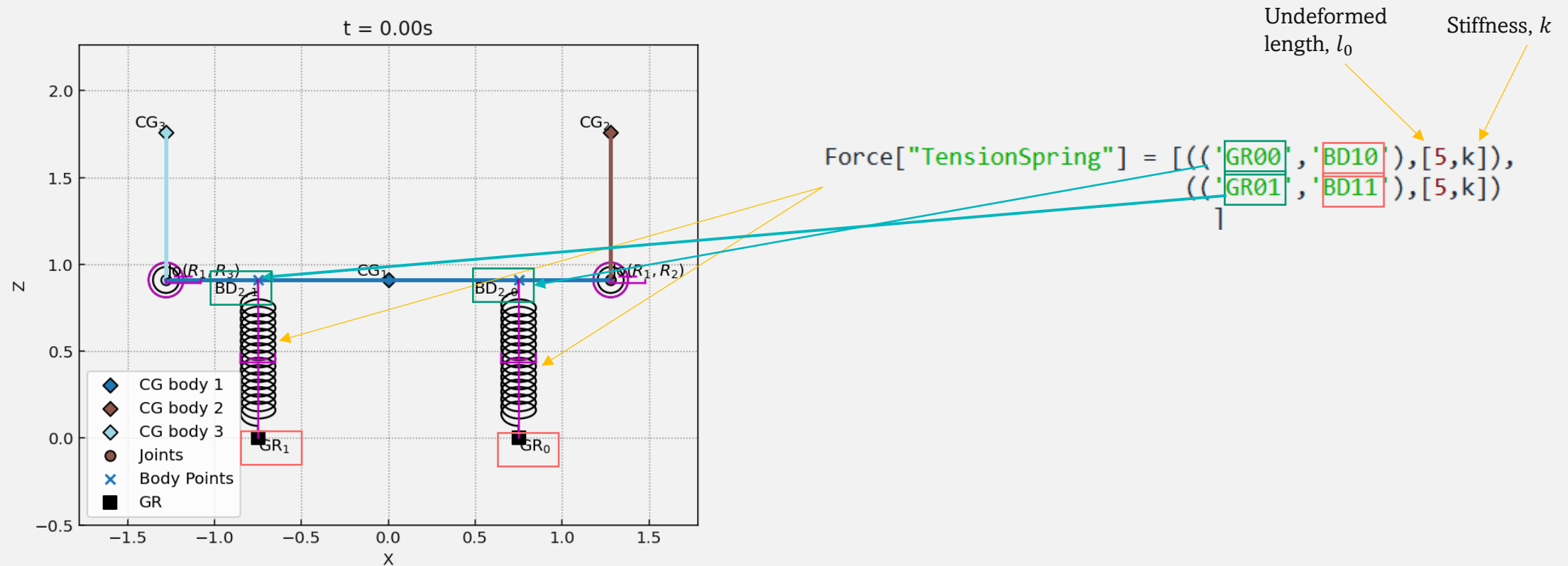


Initial_Points["GR"] = $[[0.75, 0], [-0.75, 0]]$

Initial_Points["BD"] = {}
Initial_Points["BD"][1] = $[[0.75, 0], [-0.75, 0]]$

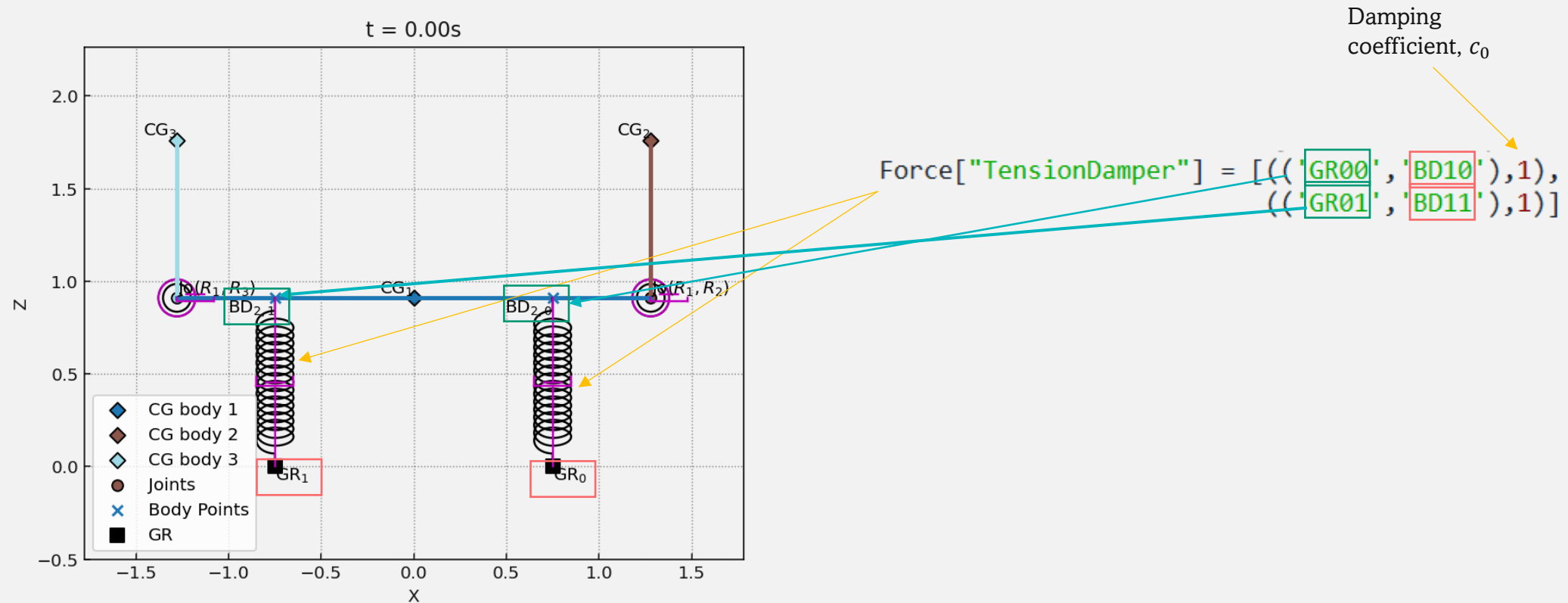
Go to next slide to continue animation

Forces definition: tension springs



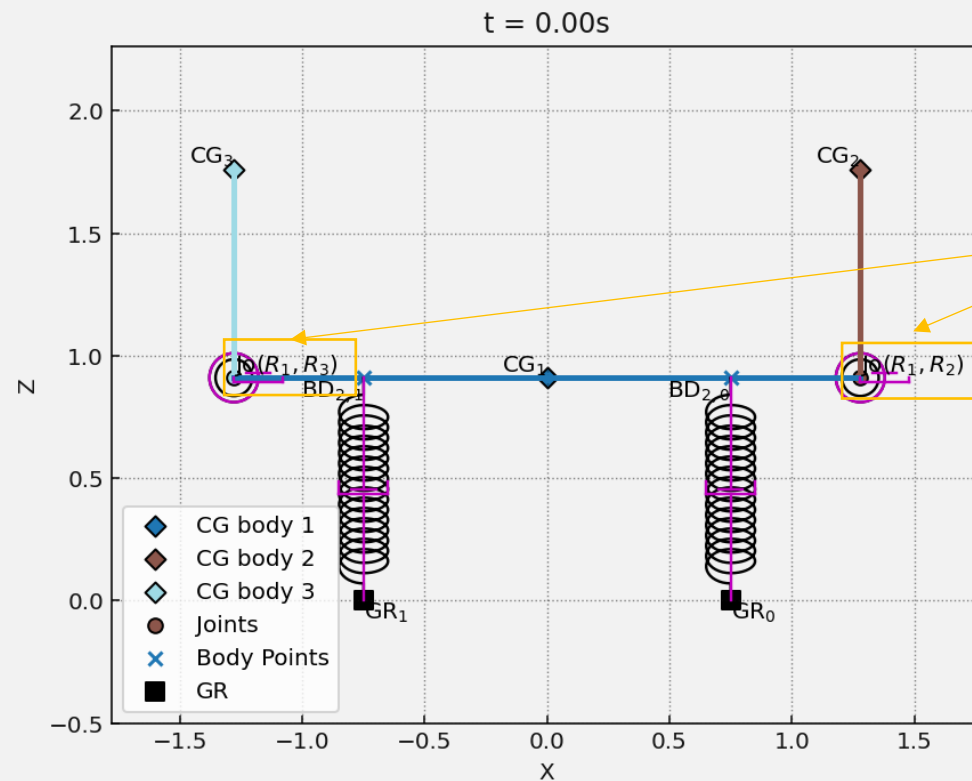
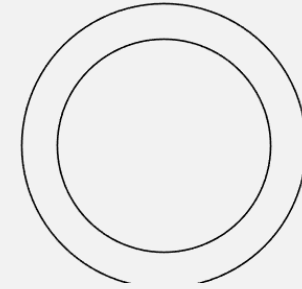
Go to next slide to continue animation

Forces definition: tension dampers



Go to next slide to continue animation

Forces definition: torsion springs



Force["TorsionSpring"] = [((1,2), [0,10]),
((1,3), [0,10])]

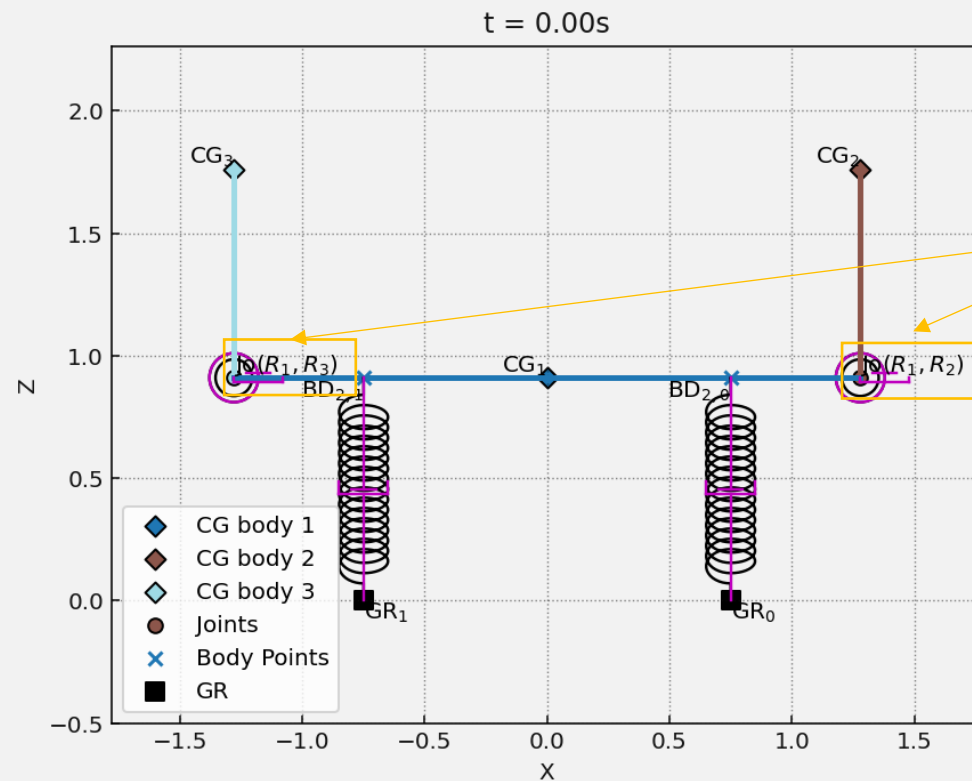
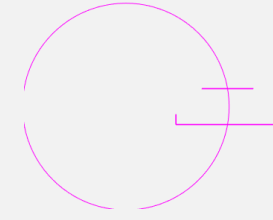
Undeformed angle, θ_0

Stiffness, k_θ

Bodies connected at joint

Go to next slide to continue animation

Forces definition: torsion dampers



Force["TorsionDamper"] = [((1,2), 0), ((1,3), 0)]

Bodies connected at joint

Damping coefficient, c_θ

Go to next slide to continue animation